

## Arduino founder plans 'sustainable' growth

Massimo Banzi, the developer of the Arduino board, has agreed to acquire 100 per cent ownership of Arduino AG, the company that owns all Arduino trademarks. The latest development is supposed to help the company generate sustainable growth through its open source hardware and software developments.

"This is the beginning of a new era for Arduino in which we will strengthen and renew our commitment to open source hardware and software, while in parallel setting the company on a sound financial course of sustainable growth," said Banzi, in an official statement.

As a result of the acquisition, Banzi, 49, has become the new chairman and CTO of Arduino. The CEO, Federico Musto, has also been replaced by Dr Fabio Violante. "In the past two years, we have worked very hard to get to this point. We envision a future in which Arduino will apply its winning recipe to democratise the Internet of Things for individuals, educators, professionals and businesses," said Dr Violante.

Developed as an open source project back in 2003, Arduino is aimed at providing affordable solutions to individuals to build new devices. The boards under the Arduino range are available as open hardware and are compatible with a range of sensors and actuators. Last month, the Banzi-led company even partnered with the LoRa Alliance to start building hardware with the LoRaWAN standard.

In May this year, the Arduino Foundation began to build an open source ecosystem for sectors like education, IoT markets, makers and receivers. "Our vision remains to continue enabling anybody to innovate with electronics for a long time to come," said Banzi.

## PayPal launches Technology Innovation Labs in India

PayPal has launched two of its Technology Innovation Labs in India to support developments specific to new age technology. Located at PayPal's Chennai and



Bengaluru centres, the labs are the first in India, opened after the Palo Alto firm launched its US and Singapore labs.

"India is a hotbed for technology innovation given its evolving startup ecosystem, diverse merchant profiles and enormous talent pool," said Mike Todasco, director of innovation, PayPal. "To cater to their needs in the most effective manner, we are delighted to announce the launch of our newest Technology Innovation Labs in India, where the focus will be on fuelling new age technology and giving rise to unconventional ideas with the potential to transform the ecosystem we operate in," he added.

PayPal's Technology Innovation Labs will support diverse fields including machine learning, artificial intelligence, data science, Internet of Things, penetration testing, software-defined radios and wireless communication, virtual and augmented reality, computer vision and basic robotics. The company will provide equipment like Raspberry Pi boards with sensor kits, AlphaBot kits, Amazon Echo, LeapMotion and 3D printers, among others.

"Enabling innovation and creating amazing experiences for our customers is at the heart of PayPal's global success, and the Innovation Lab is another step to foster this spirit in our development centres in India," said Guru Bhat, general manager — technology and head of engineering, PayPal.

In addition to providing relevant hardware to kickstart the innovative developments, PayPal is set to integrate its native incubation centre. Called the PayPal Incubator, the centre was launched back in 2013 with an aim to support India-origin startups.

## Google starts discriminating against poor quality Android apps



Google is all set to improve the user experience on Android by enhancing its search and discovery algorithms on Play Store. This will have a direct impact on apps that have quality issues.

A new Android vitals dashboard in the Google Play Console was revealed at I/O 2017 earlier this year. The technology is designed to understand and analyse inferior app behaviour such as excessive battery consumption, slow render times and crashes — and hence set benchmarks for what passes as a quality app.

"Developers focusing on performance can use the Play Console to help find and fix a number of quality issues," wrote Andrew Ahn, product manager of Google Play, in a blog post.

Google reports that the change in its algorithms has shown that more users have downloaded quality apps. The Android maker also recommends developers to examine ratings and reviews that they received on their apps to get additional insights about their quality.

If you are about to launch your app and test its functionality in the alpha or beta stage, you can use the pre-launch report to fix issues ahead of mass downloads. Likewise, Android vitals can be applied to identify performance issues reported by opt-in devices.